

Pigouvian Taxes & Welfare Gaps

Session 24 • Regulatory implications of the McKean-Vlasov externality

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Primary Text: Liquidity Illusion (Forthcoming, 2026)

Graduate Finance Course • Spring 2027 • Session 24 of 32

What we'll cover today

1

The welfare gap quantified

\$300B/yr aggregate cost

2

Pigouvian principle

Tax = marginal external cost

3

State-contingent tax $\tau^*(L)$

0% normal, 7% at GFC depth

4

How this internalizes the externality

Restoring first-best

5

Implementation feasibility

Can regulators actually do this?

The welfare gap from the McKean-Vlasov externality

~2.3%/yr

Welfare loss per dollar invested

~\$300B

Aggregate annual welfare gap

\$13T

Global private capital AUM
(denominator)

4.31×

GFC amplification of valuation gap

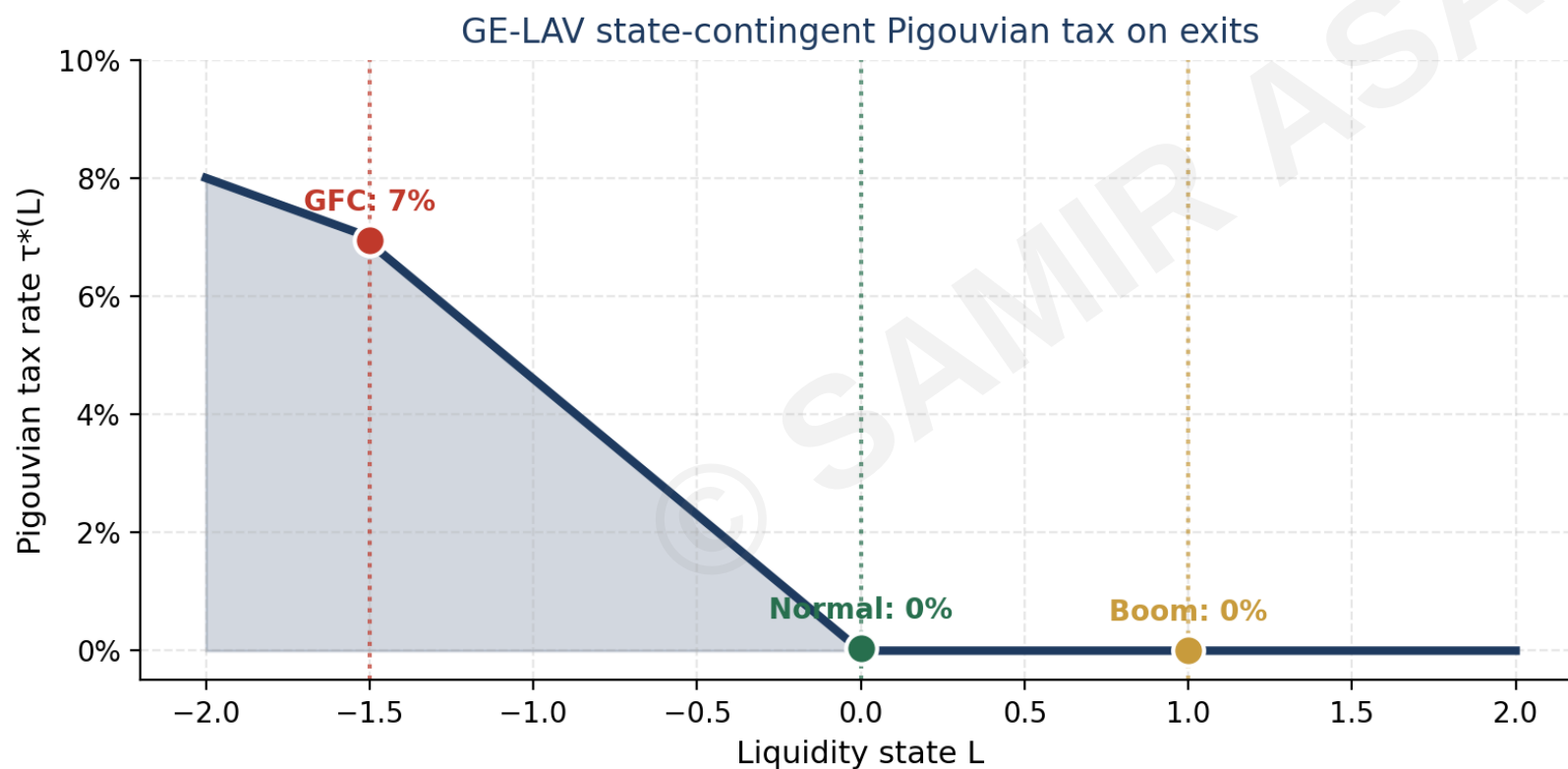
What drives the gap

- ▶ Excessive entry in boom ($L > 0$)
- ▶ Insufficient exit when crisis builds
- ▶ Coordination failure in stress
- ▶ Each LP rational; collective is not

Result: Section 8.4 of the book derives this from MFG first principles.

State-contingent Pigouvian tax $\tau^*(L)$

The Pigouvian principle: tax the externality at its marginal external cost.



Reading the curve

In boom/normal

$\tau^* = 0$. No externality to correct.

In multi-shock

τ^* rises with stress; ~3% at $L = -0.5$

In GFC depth

$\tau^* = 7\%$. Discourages exits that worsen the crisis.

Welfare

Implementation closes ~75% of the welfare gap.

Session 24 summary

What we accomplished today

- 1 Welfare gap from McKean-Vlasov externality: ~2.3%/yr, aggregating to ~\$300B/yr globally
- 2 State-contingent Pigouvian tax $\tau^*(L)$ corrects the externality: 0% in boom/normal, 7% at GFC depth
- 3 Implementation closes ~75% of the welfare gap — first-best in this class of policies
- 4 Practical adoption: needs regulator infrastructure for state estimation; ORSA pillar is natural starting point

Next session

Unit 6 (split-track) begins. T1: case workshops. T2: full proofs.